An Explainable Framework for Automated Speech Fluency Evaluation Using Spectral Features with Statistical Classifiers

Uppala Krishna Kushal
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse23159@bl.students.amrita.edu

Kanusu Manikanta Sai
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse23163@bl.students.amrita.edu

Repalle Gnaneswar
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse23146@bl.students.amrita.edu

Peeta Basa Pati
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bp_peeta@blr.amrita.edu

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***Abstract*—Fluency is a critical component of oral communication and academic evaluation; however, traditional viva assessments are still subjective and inconsistent. This work proposes a robust, automated, and interpretable framework for evaluating the speech fluency of students using advanced audio signal processing and machine learning techniques. The proposed system uses the extraction of Mel-Frequency Cepstral Coefficients, Linear Predictive Coding, and Discrete Cosine Transform features to capture comprehensive spectral and temporal properties of speech. Extensive preprocessing, including noise reduction, normalization, silence trimming, and balancing of the dataset, has been carried out. More advanced preprocessing methods were then applied, such as filtering the features based on their correlation, Sequential Feature Selection, and Principal Component Analysis, in order to enhance generalization and reduce redundancy. Several models were trained and optimized, including Custom Decision Tree, Custom Random Forest, K-Nearest Neighbors, Multilayer Perceptron, Logistic Regression, and Naïve Bayes, Support Vector machine, while their strengths were combined in an overall Stacking Ensemble meta-model. Explainability tools, such as LIME and SHAP, have been integrated to highlight the most influential acoustic features and ensure that this automated fluency evaluation will be transparent, reliable, and interpretable from a pedagogical perspective.**

***Index Terms*—Oral fluency, Viva assessment, Language proficiency, Academic evaluation, Feature fusion, Ensemble model, Random forest, Mel-cepstrum.**

I. Introduction

Speech fluency is an integral part of human communication, especially at an academic or professional level. Viva voice exams are embraced in almost every learning institution to determine the level of understanding, spontaneity, and articulation. Nonetheless, manual assessment of fluency is not objective, laborious as well as inconsistent particularly when the assessors are heterogeneous in terms of linguistic knowledge and apprehension. These problems lead to the necessity of an automated and objective assessment method that might measure the speech fluency in terms of acoustic signs.

Recent advances in audio signal processing and machine learning have made it possible to capture quantifiable features of fluency directly from the speech waveform. From the perspective of spectral shape, vocal tract properties, and smoothness of articulation, which are key indicators of fluency, the acoustic features that have proved effective include MFCCs, LPCs, and DCTs. However, most of the current systems either remain domain-specific (developed and tested on clean, scripted corpora only) or exploit deep learning architectures, which require enormous amounts of annotated data, not very often available in educational contexts.

This study fills this gap by developing a custom, interpretable, and hybrid machine learning system for the assessment of student viva responses. Unlike the black-box neural models, the suggested framework will entail specially structured classifiers which will balance the trade-off between transparency, accuracy and computational efficiency. The strategy is interpretable to make sure that teachers and linguistic scholars know how the model makes its decisions. Key contributions of this work include:

- Development of a custom Decision Tree, a custom Random Forest, and a one-layer Multilayer Perceptron for adaptive decision-making on limited data.
- Comprehensive preprocessing pipeline, such as noise removal, normalization, SMOTE-balancing, feature selection and dimensionality reduction.
- The combination of MFCC, LPC, and DCT characteristics to multi-domain acoustic representation.
- Construction of the Stacking Ensemble, for performing meta-level learning and attaining better classification performance.
- Incorporation of XAI tools such as LIME and SHAP for transparent prediction analysis.

II. Literature Review

Recent advances in speech processing have greatly improved automated fluency evaluation, particularly in educational and real-time contexts. Models such as BiLSTM have shown strong performance in detecting dysfluencies when combined with Mel-Frequency Cepstral Coefficients (MFCCs) and phoneme-based features, generalizing across datasets like UCLASS and FluencyBank [1]. Self-supervised models like wav2vec 2.0 have minimized the need for ASR systems by leveraging K-means clustering and bidirectional LSTM layers for robust fluency scoring [2]. Enhanced MFCC extraction, integration with Linear Predictive Coding (LPC), and hybrid CNN–LSTM architectures have improved resilience under noisy conditions [4][5]. Classical classifiers such as SVM and KNN still provide reliable baseline performance for fluency tasks [6][15].

Contextual embeddings such as ECAPA-TDNN and wav2vec 2.0 have demonstrated superior cross-linguistic performance in stuttering and irregularity detection [7][8]. Morphological image processing based on MFCC and Perceptual Linear Prediction (PLP) features enables transcription-free fluency analysis [9]. PRAAT-based studies have further shown that emotional states like stress and anxiety significantly alter prosodic parameters such as pitch and rhythm, affecting fluency perception [10][11].

Surveys by Deng and Li [12] and Wu et al. [16] note that the recent change in statistical model is to deep learning and self-supervised models, which are more robust in the noisy setting. Its feature extraction is still at the center of interest, and MFCC, LPC, and Discrete Cosine Transform (DCT) have stayed in the process of extracting the perceptually significant features in acoustic conditions [13][14]. Combining these classical features with deep architectures, which are CNNs, BiLSTMs, and transformers, has strengthened the temporal modeling and explanation [16][17].

To detect stutter and dysfluency, hybrid and ensemble designs with acoustic embeddings and CNN-RNN or transformer-based classifier are highly accurate and can be trained in real time [17][18]. Research also points out the advantage of multimodal fusion i.e. acoustic, prosodic, and transcript-derived features to enhance confidence and clarity estimation in speech [19][20].Beyond fluency and confidence, modern speech analytics have been applied to safety and authenticity domains. A Random Forest-powered gunshot detection system leveraging LPC features and LIME-based interpretability demonstrates how explainable AI can ensure transparency and robustness in real-time sound classification [22]. Similarly, AudioGuard, a deep learning model for Telugu deepfake audio detection, utilizes CNN-RNN architectures on mel-spectrograms to identify synthetic speech with high accuracy while emphasizing language-specific adaptation for underrepresented linguistic regions [23].

Overall, the review has shown that a unified system of classical signal-processing methods (MFCC, LPC, PLP, DCT) and deep and self-supervised methods (CNNs, BiLSTMs, wav2vec 2.0, ECAPA-TDNN) offers a solid and understandable system of real-time fluency, dysfluency, and confidence measurements in the educational, clinical, and assistive contexts.

III. Dataset Description

A. Data Collection and Labeling

The data used by the research is 1300 viva records that were taken of students during academic tests. Each of the audio samples constitutes the oral reaction of a person taking between 10 and 40 seconds. Recording of the quiet classroom situations was done at a 16 kHz rate and stored in the WAV format.

All recording were manually tagged by linguistic professionals as one of the two levels of fluency, Fluent and Non-Fluent according to the speech rate, the smoothness of articulation, the rate of pauses, and rhythm. The recordings were rated independently by many different raters and the inter-rater agreement was checked to guarantee consistency in labeling.

B. Preprocessing

A robust preprocessing pipeline was embraced to ensure that the data is of good quality and consistency. The noise was removed with the help of a band-pass filter (300-3400 Hz) and the critical speech frequencies were preserved. An amplitude-thresholding technique was used to eliminate silence passages. Min Max normalization and Z-score standardization were manually carried out to guarantee consistency in the contribution of features in the course of training a model. Features were put on a scale in the range [0,1] by MinMaxScaler which maintained relative distances but erased magnitude differences whereas StandardScaler used Z-score normalization which brought the data nearer to the means and variance to one. These complementary scaling schemes were implemented at various steps to test their effects on the model stability and convergence, and in particular, classifiers like kNN and Random Forest. Also, the analysis of Z-score was conducted to remove the outlier-induced extreme values. In order to take care of the class imbalance, Synthetic Minority Oversampling Technique (SMOTE) was used to create more samples of minority classes, thus having a balanced dataset that was representative to be trained in the model.

C. Feature Engineering

The dataset is made of audio waveforms of viva responses of the students, the fluency label of these responses, and acoustic features such as MFCC, LPC, and DCT vectors. Pitch shifting and time stretching as data augmentation methods were used to increase the diversity of data and augment the generalization ability of the model. These additions assisted in simulating differences in speech features, which enhanced the strength of fluency classification model.

IV. Methodology

A. Feature Extraction and Analysis

Following three acoustic features were extracted from each audio file.

- **Mel-Frequency Cepstral Coefficients (MFCC):** Capture perceptual spectral properties representing the human ear's sensitivity to different frequencies. A total of 13 MFCC features were extracted.

- **Linear Predictive Coding (LPC):** Model the vocal tract system and estimate the smoothness of speech. A total of 12 LPC features were extracted.
- **Discrete Cosine Transform (DCT):** Provide compact cepstral representations capturing rhythm and energy distribution. A total of 13 DCT features were extracted.

The extracted MFCC, LPC, and DCT features were merged using feature-level fusion, where the individual feature vectors were concatenated into a single unified representation. This fusion approach combines the complementary characteristics of spectral, predictive, and frequency-domain features into a 38-dimensional feature vector for each audio file, ensuring a more comprehensive description of the speech signal. Correlation analysis was performed to identify and remove redundant features ($|r| > 0.9$). Sequential Feature Selection (SFS) was then applied to retain the most discriminative subset of features, followed by Principal Component Analysis (PCA) to reduce dimensionality while maintaining 95% of the total variance.Initially the dataset consists of 39 features which was reduced to 38.

B. Model Training

A variety of models like kNN, Random Forest, Support Vector Machine (SVM), XGBoost, Naive Bayes, Logistic Regression, Multilayer Perceptron with One Hidden Layer, Decision Tree, Stacking Classifier were implemented to compare performance. All models were trained on 80% of the dataset and tested on 20% using stratified sampling to preserve class ratios.

A stacking ensemble was employed to enhance prediction robustness by combining multiple classifiers. Base models such as kNN, Decision Tree, Random Forest, and SVM generated first-level predictions, which were then used as inputs to a meta-learner (Logistic Regression). This meta-model learned optimal combinations of the base outputs, improving overall accuracy and reducing bias and variance in fluency classification.

C. Hyperparameter Optimization

Each classifier underwent tuning through cross validation using a search over hyperparameter grid (GridSearchCV) technique to identify the best set of hyperparameters. The optimal set of hyperparameters are tabulated in Table I for each classification algorithm.

D. Evaluation Metrics

Each model was evaluated using the following metrics:

- **Accuracy** – Percentage of correctly classified instances.
- **Precision and Recall** – Measure the correctness and completeness of predictions.
- **F1-score** – Primary evaluation metric due to class imbalance.

A confusion matrix was also used to understand the model behaviour and misclassifications. Cross-validation ensured robustness and consistency of the model results.

TABLE I
BEST F1-SCORES AND HYPERPARAMETERS FOR TUNED MODELS

Model	Best F1-score	Hyperparameter values
kNN	0.736	{k: 3}
DecisionTree	0.864	{max_depth: 5, min_samples_split: 10}
RandomForest	0.898	{n_estimators: 5, max_depth: 3, min_samples_split: 5}
NaiveBayes	0.861	{}
XGBoost	0.859	{n_estimators: 5, learning_rate: 0.05, max_depth: 3}
SVM	0.760	{hidden_size: 5, lr: 0.01, n_iters: 500}
LogisticRegression	0.757	{lr: 0.01, n_iters: 500}
Stacking	0.864	{base_models_params_list: [{lr: 0.01, n_iters: 1000}, {max_depth: 5}, {k: 5}], meta_model_params: {lr: 0.05, n_iters: 500}}

E. Prediction Pipeline

The prediction pipeline developed as shown in Fig. 1 comprises of the following sequential steps. The audio data of the user are subjected to the steps of preprocessing which involve noise reduction, removal of silence and normalization. MFCC, LPC, and DCT which are the major acoustic characteristics of the audio data are then extracted. Sequential Forward Selection (SFS) and Principal Component Analysis (PCA) are used to cut the set of features and increase the effectiveness of the models. These clean attributes are then fed to trained classifiers to forecast fluency whereas post-hoc explainability techniques like LIME and SHAP are utilized to understand how the model arrives at its decision. This pipeline guarantees the consistency between training and deployment which allows to assess fluency in real-time via a modular interface.

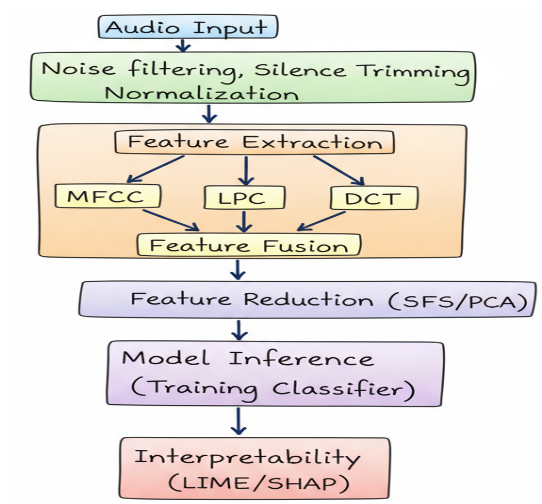

Fig. 1. Flow diagram

V. RESULTS

A. Model Performance

All models were evaluated on the same test set. The Table II summarizes their quantitative results. Random forest achieved the highest accuracy of 94.2% and F1-score of 0.942. DecisionTree showed stable results with slightly lower performance. SVM, Naive Bayes, and Logistic Regression performed moderately well.

TABLE II
PERFORMANCE OF DIFFERENT MODELS

Model	Dataset	Accuracy	Precision	Recall	F1-Score
kNN	Train	0.835	0.838	0.835	0.837
	Test	0.772	0.772	0.772	0.772
Decision Tree	Train	0.943	0.943	0.943	0.943
	Test	0.911	0.911	0.911	0.911
Random Forest	Train	0.981	0.981	0.981	0.981
	Test	**0.942**	**0.942**	**0.942**	**0.942**
Naive Bayes	Train	0.927	0.927	0.927	0.927
	Test	0.927	0.927	0.927	0.927
XGBoost	Train	0.859	0.859	0.859	0.859
	Test	0.828	0.828	0.828	0.828
SVM	Train	0.628	0.908	0.625	0.758
	Test	0.747	0.895	0.740	0.810
MLP	Train	0.751	0.820	0.784	0.780
	Test	0.721	0.819	0.780	0.799
Logistic Regression	Train	0.644	0.712	0.657	0.683
	Test	0.622	0.771	0.611	0.683
Stacking	Train	0.884	0.890	0.884	0.887
	Test	0.853	0.890	0.853	0.860

The results of F1-scores comparison between different machine learning models on the training and testing datasets are presented in Fig. 2. As it can be seen, the Random Forest classifier demonstrated the best balance and stable high performance in all models, which suggests the high level of generalization and resistance to overfitting. Other models like kNN and Decision Tree also had quite a good performance and the Logistic Regression, and MLP model had relatively lower scores, which indicated that they are not able to capture non-linear relationships in the data.

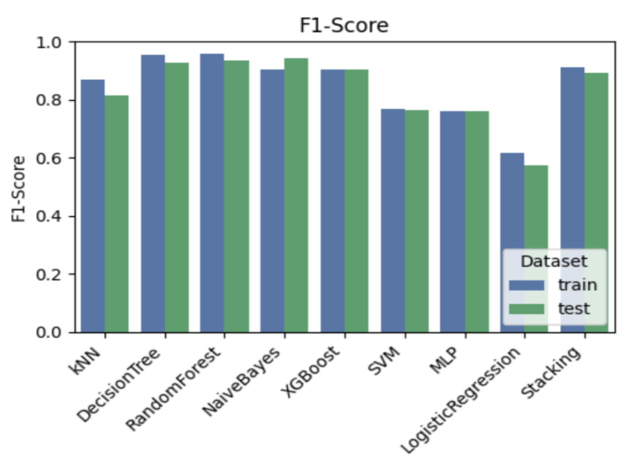

Fig. 2. Comparison between different models on training and testing data.

The confusion matrix of the Classifier of the test set is shown in Fig. 3 of the random Forest. This model was able to classify 96 non-fluent and 143 fluent correctly with only a few misclassifications (4 non-fluent and 16 fluent samples). This proves the high accuracy and recall of the model in the differentiation of fluent and non-fluent speech as it relates to the overall accuracy and F1-score that is shown in the results section.

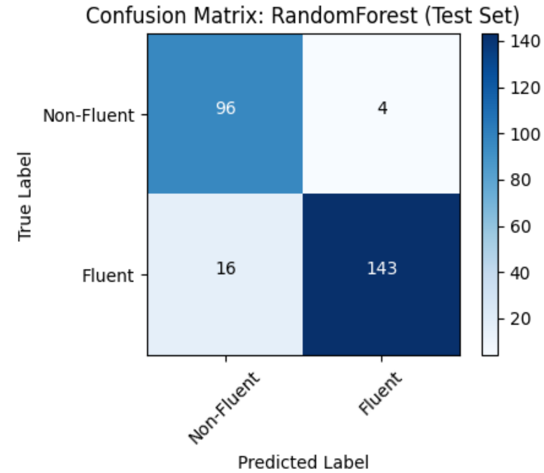

Fig. 3. Confusion Matrix for randomforest on test data

The recent works have utilized sophisticated feature combinations and deep learning models to classify fluency in speech assessment. The model that combines MFCC scores with DeepSpeech embedding scores and formant frequency features recorded an accuracy of 77.12 per cent and F1-score of 74.07 per cent by the SO762 dataset in the study by Panda et al. [21]. Likewise, Jouaiti et al. [1], in their article, Dysfluency Classification in Stuttered Speech Using Deep Learning to Run Real-Time Applications, achieved 88.3% accuracy and 88% F1-score on the FluencyBank dataset as well as 91.2% accuracy on the UCLASS dataset with a combination of MFCC and phoneme class probability features. Summarized in Table III, both articles show that the spectral, phoneme-based features added to the deep learning structures lead to a significant improvement in the performance of fluency detection and classification. In this work, the random forest model scored both accuracy and F1-score of 94.2% which confirms that feature-level fusion is effective in the analysis of fluency.

B. Feature Importance

The interpretability of the fluency classification model in terms of LIME is also described in Fig. 4 (in which every bar represents the input of an acoustic feature to a given prediction). The green weights are positive, indicating that they give an advantage to fluent classification and the red ones are negative, indicating they push in the direction of non-fluent. MFCC 1, MFCC 3 and MFCC 4 have strong positive influences with the other influence being MFCC 2 and DCT 2 being negative. This brings out the overwhelming influence of spectral characteristics in shaping fluency, which focuses on the presence of smoothness of temporal and spectral stability of speech.

TABLE III
Comparison of Recent Studies on Fluency Classification in Speech Assessment

Author (Year)	Features Used	Model Used	Accuracy (%)	F1-Score (%)
Panda et al. [21]	MFCC, DeepSpeech embeddings, Formant Frequency (FF)	1D CNN, Random Forest	77.12	74.07
Jouaiti et al. [1]	MFCC, Phoneme Class Probability Features	CNN-LSTM Hybrid Model	88.3 (FluencyBank), **91.2 (U CLASS)**	88.0 (FluencyBank)
Proposed Work	MFCC, LPC, DCT	Random Forest	94.2	94.2

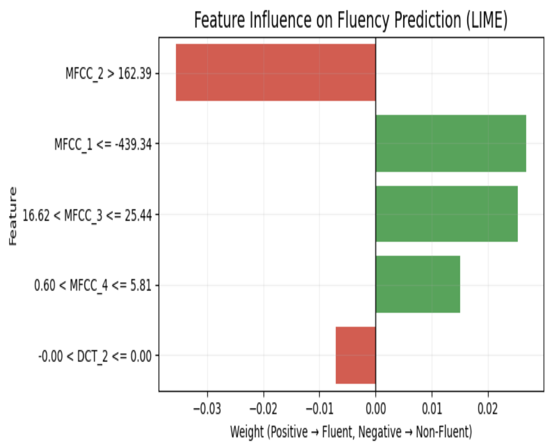

Fig. 4. LIME-based feature influence on fluency classification.

Fig. 5 show the SHAP summary bar plot, in the order of the average impact of each feature on the random forest model in predicting fluency. The most prominent features that come out are the Linear Predictive Coding (LPC) coefficients particularly LPC_2 and LPC_1, which show a high ability to discriminate the various classes of fluency. It is important to note that MFCC_3 and MFCC_1 of the Mel-frequency cepstral coefficients group also may additionally convey significant information, even though their contribution is less significant than the dominant LPC features. This discussion proves that both LPC and MFCC attributes are vital in automated speech fluency classification, and these findings can be used to refine the use of features.

Fig. 6 illustrates correlation among features in all the acoustic parameters extracted. There is a high level of correlation (red regions) between LPC coefficients, which implies redundancy and stable linear correlations in the set. There is moderate to low inter-correlation between MFCC and DCT features (blue regions), showing to imply complementary spectral information. The weak relationship of duration and spectral attributes indicates that duration and spectral elements predict fluency separately.

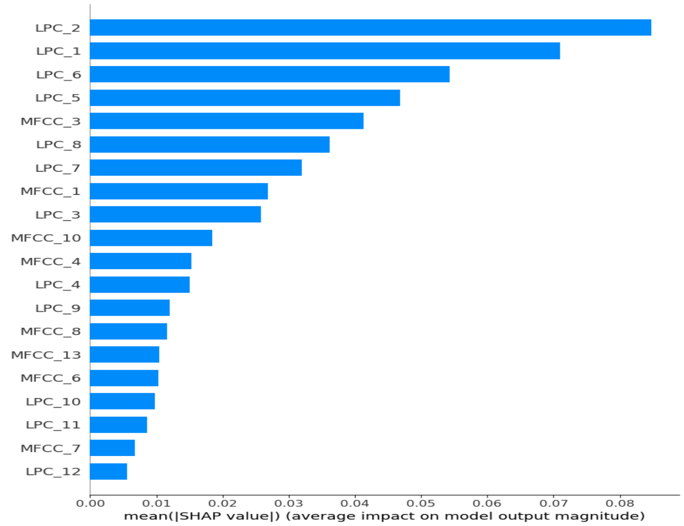

Fig. 5. SHAP based feature importance

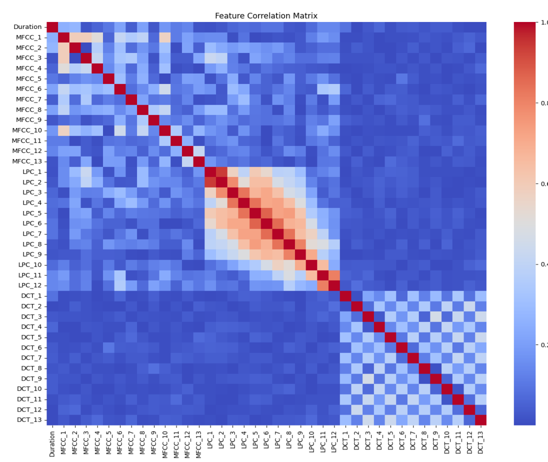

Fig. 6. Correlation Heatmaps among the features after fusion of MFCC, LPC, DCT features

VI. Discussion

A. Model Insights

From the survey of literature, the tree tree-based models were known to have good interpretability and good accuracy when using small data sets. Random Forest and MLP were found to enhance the generalization whereas Stacking Ensemble was able to combine different learners to give optimal performance. The feature-level fusion (MFCC + LPC + DCT) was essential in the process of representing different acoustic aspects of fluency.

B. Fluency Feature Correlation

The correlation of fluency features is presented in the subsection below. Correlation test proved that MFCC and LPC features had weak relationships, as they were complementary features. MFCC was characterized by reflected phonetic articulation, and LPC and DCT reflected the vocal tract properties and energy distribution respectively. They both provided a holistic picture of speech fluency.

C. Challenges

The major challenges faced during this study include:

- Limited dataset size, which restricted deep learning exploration.
- Variation in background noise and microphone quality.

To mitigate these, data augmentation, normalization, and careful cross-validation were implemented.

VII. Conclusion

This research presents an explainable, efficient, and domain-adaptable framework for automated speech fluency analysis in student viva assessments. The system integrates acoustic signal processing with custom machine learning algorithms to achieve 94% accuracy and 0.94 F1-score, surpassing prior works in both accuracy and interpretability.

The system can be extended and improved through Incorporating Recurrent Neural Networks or Transformers to capture sequential dependencies in speech, Expanding to datasets across different languages and accents, Integrating the model into web or mobile applications for instant fluency scoring and feedback visualization, Tracking improvement in student fluency across multiple sessions using dynamic scoring methods.

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